

Vibe Coding Lab 4 – Chat History

Math Flashcards Arena

Student: Hemanth V

Repository: 61_math_flashcards

Task 1 – Fix Arithmetic Calculation Bug

Vibe Coding Prompt

Fix the broken arithmetic answer calculation in the existing Pygame math flashcard game. The expected answer must correctly evaluate addition, subtraction, and multiplication according to the current operator. Preserve the existing game, input, score, and attempt functionality.

Changes Implemented

- Replaced the incorrect expected-answer calculation.
- Added correct operator-based calculations for addition, subtraction, and multiplication.
- Preserved the existing input, score, attempt, and feedback functionality.

Test Results

- Ran the corrected game.
- Tested correct arithmetic answers and confirmed they were accepted.
- Confirmed the calculations worked correctly.

Git Commit

a38df09 – Fix arithmetic calculation bug

Task 2 – Add Per-Question Timer

Vibe Coding Prompt

Add a per-question timer bar to this Pygame math flashcard game. Each question should have a 10-second countdown displayed as a visible progress bar below the active question card. The timer must reset whenever a new question is generated. If the timer reaches zero, automatically register the question as a missed attempt, show a timeout message, and generate a new question. Preserve all existing answer, score, and attempt functionality.

Changes Implemented

- Added a 10-second timer for each question.
- Added a visible timer progress bar and countdown.
- Reset the timer whenever a new question is generated.
- Timeout registers a missed attempt and generates a new question.

Test Results

- Ran the game and confirmed the countdown was visible.
- Confirmed the timer progress bar worked.
- Confirmed timeout registered a missed attempt and generated a new question.
- Confirmed existing answer and score functionality remained working.

Git Commit

92aa37f – Add per-question timer

Task 3 – Add Streak Multiplier

Vibe Coding Prompt

Add a consecutive-correct streak multiplier to the existing math flashcard game. Keep the current timer functionality unchanged. Track consecutive correct answers and increase the score reward based on the streak: 1x for the first correct answer, 2x for the second, 3x for the third, and so on. A wrong answer or timeout must reset the streak and multiplier back to the baseline. Display the current streak and multiplier in the game HUD. Preserve all existing functionality.

Changes Implemented

- Added consecutive-correct streak tracking.
- Added a multiplier that increases with each consecutive correct answer.
- Wrong answers reset the streak and multiplier to the baseline.
- Timeouts reset the streak and multiplier.
- Displayed the current streak and multiplier in the HUD.

Test Results

- Confirmed a streak reached 4 with Multiplier x4.
- Confirmed the score reward increased with the multiplier.
- Tested a wrong answer and confirmed the streak reset to 0 and multiplier to x1.
- Confirmed the timer continued working.

Git Commit

```
44cc2ab - Add streak multiplier
```

Task 4 – Add Clean Division Operator

Vibe Coding Prompt

Add a division operator to the existing math flashcard game. Keep all existing addition, subtraction, multiplication, timer, streak, and multiplier functionality unchanged. Division questions must always produce clean integer quotients with no remainder. Generate division problems so that the dividend is evenly divisible by the divisor, and calculate the expected answer as a whole number. Include '/' as a possible operator when generating questions.

Changes Implemented

- Added '/' as a possible arithmetic operator.
- Generated division problems with exact whole-number quotients.
- Used integer division for the expected answer.
- Preserved addition, subtraction, multiplication, timer, streak, multiplier, scoring, and feedback.
- Adjusted the timer text position to remove overlap with the input area.

Test Results

- Confirmed a division question appeared, such as $30 / 6$.
- Confirmed the clean answer was 5 with no remainder.
- Confirmed the timer continued working.
- Confirmed timeout reset the streak and multiplier.
- Confirmed normal addition and wrong-answer feedback still worked.
- Recorded the final AFTER video.

Git Commit

```
f2d1b41 - Add clean division operator
```